COIMBATORE INSTITUTE OF TECHNOLOGY

GOVERNMENT AIDED AUTONOMOUS INSTITUTION
AFFILIATED TO ANNA UNIVERSITY, CHENNAI

DEPARTMENT OF COMPUTING
M.Sc DATA SCIENCE

Presents

DATATHON

CREATIVITY MEETS TECHNOLOGY , IDEAS INTO SOLUTIONS

PROBLEM STATEMENTS

14th-15th MARCH

JOIN US FOR A COSMIC EXPERIENCE UNLIKE ANY OTHER.

NOTE: Expected deliverables and submission requirements will be shared on the day of the hackathon.

PS-D1

Early childhood education programs have hit a complexity wall where manual management of multi-modal, multilingual resources is no longer humanly possible. Traditional systems fail to bridge the gap between fragmented data (voice notes, videos, docs) and the specific, real-time needs of diverse stakeholders across regions. This creates a critical intelligence bottleneck that requires an autonomous layerto synthesize, translate, and proactively deliver the right context to the right person.

PS-D2

Property listing platforms have reached a labeling breaking point where the massive influx of unorganized, multi-perspective room images has outpaced human and basic algorithmic categorization. Traditional static filters fail to distinguish subtle architectural nuances, regional design styles, or "cluttered" real-world conditions, leading to massive search friction and lost revenue. This creates a systemic visual data bottleneck that requires an autonomous layer capable of independent spatial reasoning to instantly identify, high-grade, and strategically map images to the correct user intent without human oversight.

PS-D3

Sellers have hit a competitive intelligence wall where the sheer velocity of cross-platform pricing shifts, sentiment nuances in thousands of reviews, and shifting competitor tactics has outpaced human analytical capacity. Traditional data scraping fails because it lacks the autonomous reasoning to connect disparate signals—like a specific material complaint in a reviewto a pricing drop in a rival'scatalog. This creates a strategic blind spot that requiresan agentic layerto independently monitor,synthesize, and proactively suggest pivot strategies before market opportunities evaporate.

PS-D4

The US healthcare reimbursement landscape has reached a state of regulatory entropy, where the high frequency of fragmented updates across thousands of disparate portals makes manual tracking a systemic liability. Static monitoring tools fail because they cannot interpret the clinical intent of a policy change or map it to a specific billing code's financial impact. This creates a compliance-risk bottleneck that necessitates an autonomous agent capable of continuous environmental scanning, legal reasoning, and real-time execution of workflow adjustments to prevent catastrophic claim rejections.

PS-D5

Energy utility providers have hit a synchronization wall where the surge in volatile renewable inputs (solar, wind) and fluctuating electric vehicle (EV) loads has outpaced the speed of traditional grid management. Current SCADA systems are rule-bound and reactive, capable only of flagging a voltage surge or frequency drop after it threatens to trigger a blackout. This creates an operational fragility where manual dispatchers cannot process thousands of sub-second sensor signals from "prosumers" and microgrids simultaneously. Resolving this requiresan autonomous layer capable of real-time load-balancing orchestration, independent "self-healing" rerouting of power duringlocal faults, and the proactive execution of pricing signals to stabilize the grid before human operators can even perceive the imbalance.